

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0447

Silver Peak 138 kV
March 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0447. The Interconnection Customer proposes a 91.5 MW wind facility interconnecting at the Silver Peak 138 kV substation in the Pennsylvania area. The request was received on March 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 265 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,608,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WALNUT CRK 138-1 kV XFMR
Contingency: CORONET - FT HUMBUG 500 kV
Loading: 244.4% (Rating: 324 MVA)
DFAX: 5.4%
Pre-project loading: 242.9%
2. Facility: WALNUT CRK 345-1 kV Transformer Bank
Contingency: CORONET - FORT HUMBUG 500 kV
Loading: 167.1% (Rating: 474 MVA)
DFAX: 5.4%
Pre-project loading: 166.0%
3. Facility: BIG HOLLOW RD - WEST CAMPUS 138 kV
Contingency: EDES TAP 2 - W CAMPUS 345 kV Line
Loading: 144.0% (Rating: 249 MVA)
DFAX: 5.2%
Pre-project loading: 142.1%
4. Facility: LUTESVILLE - SUNRISE CTR 345 kV
Contingency: SUNRISE CTR - CAYUCOS SUBSTATION 500 kV
Loading: 117.8% (Rating: 1,202 MVA)
DFAX: 5.3%
Pre-project loading: 117.4%
5. Facility: MCADORY - LUTESVILLE 138 kV line
Contingency: SUNRISE CTR - LUTESVILLE SUBSTATION 345 kV
Loading: 108.1% (Rating: 233 MVA)
DFAX: 15.5%
Pre-project loading: 102.1%

6. Facility: LAS PULGAS - THERMO 138 kV
Contingency: 7TH AVENUE NORTH SUBSTATION - NOTTELY 138 kV L
Loading: 100.8% (Rating: 186 MVA)
DFAX: 19.9%
Pre-project loading: 91.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WALNUT CRK 138-1 kV XFMR	Replace transformer with higher-rated unit	\$4,250,000	4% (\$159,000)
WALNUT CRK 345-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$17,678,000	4% (\$661,000)
BIG HOLLOW RD - WEST CAMPUS 138 kV	Rebuild line with higher-capacity conductor	\$1,569,000	8% (\$123,000)
LUTESVILLE - SUNRISE CTR 345 kV	Rebuild line with higher-capacity conductor	\$41,218,000	1% (\$397,000)
MCADORY SUBSTATION - LUTESVILLE 138 kV	Reconductor line	\$1,718,000	42% (\$730,000)
LAS PULGAS - THERMO 138 kV	Reconductor line	\$3,209,000	48% (\$1,538,000)
	Total allocated to project		\$3,608,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.